Bragg reflection: determining the lattice constants of monocrystals

Objects of the experiment

- Investigating and comparing Bragg reflection at an LiF and an NaCl monocrystal.
- Determining the lattice constant a_0 of NaCl and LiF.

Principles

Bragg's law of reflection describes the diffraction of plane waves at a monocrystal as the selective reflection of the waves at a set of lattice planes within the crystal. Due to the periodicity of the crystal, the lattice planes of a set have a fixed spacing *d*. An incident wave with the wavelength λ is reflected with maximum intensity when the Bragg condition

$$n \cdot \lambda = 2 \cdot d \cdot \sin\vartheta \qquad \text{(I)}$$

n: diffraction order
λ: wavelength
d: spacing of lattice planes

is fulfilled (see experiment P6.3.3.1). The angle ϑ shows the direction of the incident and reflected wave with respect to the set of lattice planes and is often referred to as the glancing angle.

In a cubic crystal with NaCl structure (cf. Fig. 1), the lattice planes run parallel to the surfaces of the crystal's unit cells in the simplest case. Their spacing *d* corresponds to one half the lattice constant:

$$d = \frac{a_0}{2} \qquad \text{(II)}$$

This lets us use (I) as an equation for determining the lattice constant a_0:

$$n \cdot \lambda = a_0 \cdot \sin\vartheta \qquad \text{(III)}$$

In other words, to determine a_0 we need to measure the glancing angle ϑ for a known wavelength λ and diffraction order *n.* This method is more precise when the glancing angles are also measured in higher diffraction orders.

In this experiment, the molybdenum x-rays are used as radiation of a known wavelength. Table 1 shows its wavelengths λ.

Fig. 1 Three-dimensional representation of the structure of NaCl
d: Spacing of lattice planes in [1,0,0]-direction
a_0: lattice constant

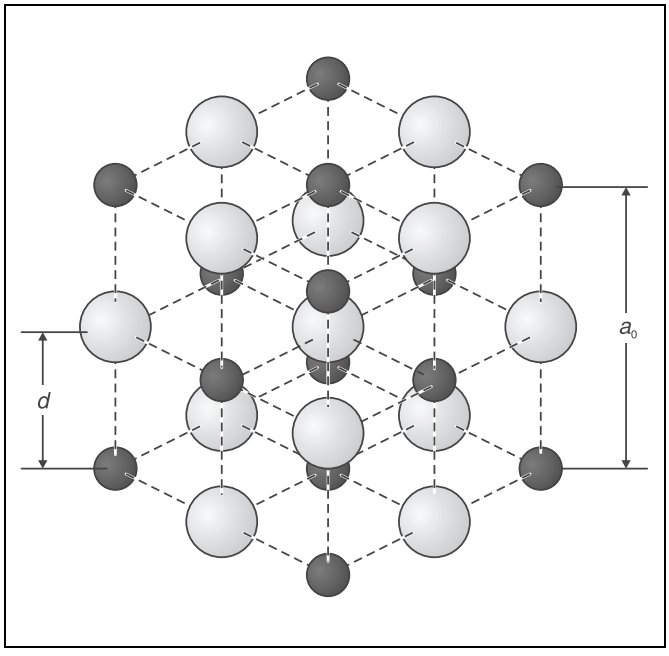


Table 1: Wavelengths of the characteristic x-ray radiation of molybdenum (weighted means [1])

Line	$\frac{\lambda}{\text{pm}}$
K_α	71.08
K_β	63.09

A Geiger-Müller counter tube is used to detect the x-rays; this instrument and the crystal are both pivoted with respect to the incident x-ray beam in 2ϑ coupling – the counter tube is turned by twice the angle of the crystal (cf. Fig. 2). The zero point $\vartheta = 0°$ is characterized by the fact that the lattice planes and the

Apparatus

1 X-ray apparatus	554 811
1 End-window counter for α, β, γ and x-ray radiation	559 01
1 LiF monocrystal for Bragg reflection	554 77

additionally required:

1 PC with Windows 9x/NT

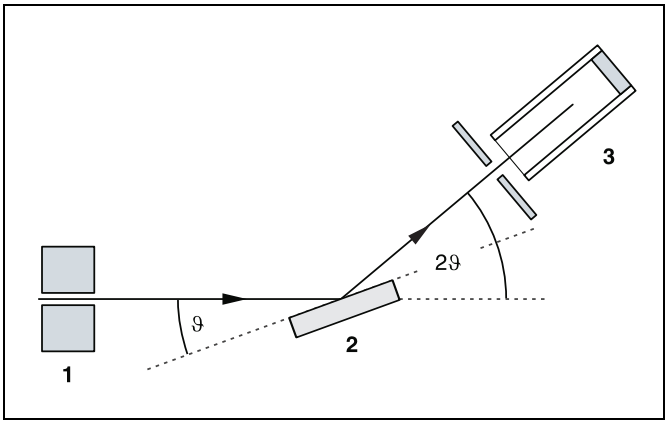


Fig. 2 Schematic diagram of diffraction of x-rays at a monocrystal and 2ϑ coupling between counter-tube angle and scattering angle (glancing angle)
1 collimator, **2** monocrystal, **3** counter tube

axis of the counter tube are parallel to the incident x-ray beam. As the lattice planes are seldom precisely parallel to the surface of the crystal, the zero point of each crystal must be calibrated individually.

Safety notes

The x-ray apparatus fulfills all regulations governing an x-ray apparatus and fully protected device for instructional use and is type approved for school use in Germany (NW 807/97 Rö).

The built-in protection and screening measures reduce the local dose rate outside of the x-ray apparatus to less than 1 μSv/h, a value which is on the order of magnitude of the natural background radiation.

- Before putting the x-ray apparatus into operation inspect it for damage and to make sure that the high voltage is shut off when the sliding doors are opened (see Instruction Sheet for x-ray apparatus).
- Keep the x-ray apparatus secure from access by unauthorized persons.

Do not allow the anode of the x-ray tube Mo to overheat.

- When switching on the x-ray apparatus, check to make sure that the ventilator in the tube chamber is turning.

The goniometer is positioned solely by electric stepper motors.

- Do not block the target arm and sensor arm of the goniometer and do not use force to move them.

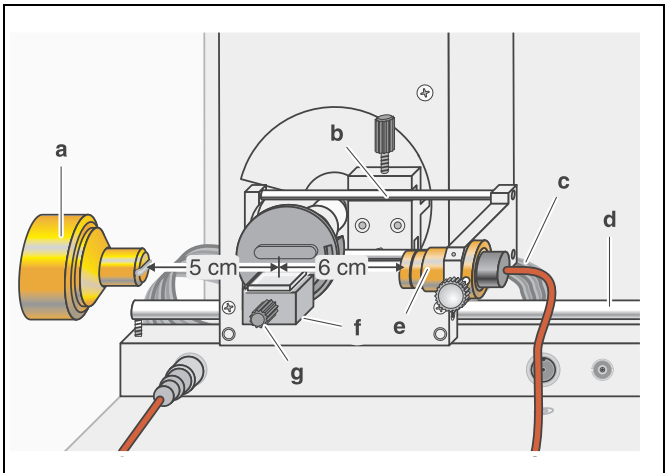
Setup

Setup in Bragg configuration:

Fig. 3 shows some important details of the experiment setup. To set up the experiment, proceed as follows (see also the Instruction Sheet for the x-ray apparatus):

– Mount the collimator in the collimator mount **(a)** (note the guide groove).
– Attach the goniometer to guide rods **(d)** so that the distance s_1 between the slit diaphragm of the collimator and the target arm is approx. 5 cm. Connect ribbon cable **(c)** for controlling the goniometer.
– Remove the protective cap of the end-window counter, place the end-window counter in sensor seat **(e)** and connect the counter tube cable to the socket marked GM TUBE.

Fig. 3 Experiment setup in Bragg configuration



– By moving the sensor holder **(b)**, set the distance s_2 between the target arm and the slit diaphragm of the sensor seat to approx. 6 cm.
– Mount the target holder **(f)** with target stage.
– Manually align the target and sensor arm horizontally using the ADJUST knob and save these positions as the "zero position of the measuring system" by pressing TARGET, COUPLED and β Limits at the same time (see Instruction Sheet for x-ray apparatus).

Preparing the PC-based measurement:

– Connect the RS–232 output and the serial interface on your PC (usually COM1 or COM2) using the 9-pin V.24 cable (supplied with x-ray apparatus).
– If necessary, install the software "X-ray Apparatus" under Windows 9x/NT (see Instruction Sheet for x-ray apparatus) and select the desired language.

Carrying out the experiment

Notes:

NaCl and LiF crystals are hygroscopic and extremely fragile. Store the crystals in a dry place; avoid mechanical stresses on the crystals; handle the crystals by the short faces only.

If the counting rate is too low, you can reduce the distance s_2 between the target and the sensor somewhat. However, the distance should not be too small, as otherwise the angular resolution of the goniometer is no longer sufficient to separate the characteristic K_α and K_β lines.

a) Bragg reflection at an LiF monocrystal:

– Loosen knurled screw **(g)**, place the LiF crystal flat on the target stage, carefully raise the target stage with crystal all the way to the stop and gently tighten the knurled screw (prevent skewing of the crystal by applying a slight pressure).
– Set the tube high voltage U = 35.0 kV and the emission current I = 1.00 mA.

Determining the zero position of the measuring system

– In coupled scanning mode, set the target to about 10.2° using the ADJUST knob.
– Switch on the tube high voltage with HV on/off.
– Leave the target position unchanged and, in sensor scanning mode, manually find the counting rate maximum for the first reflection maximum of the K_a line.
– Leave the sensor unchanged in the maximum counting-rate position and manually find the maximum of the counting rate in target mode.
– Switch between sensor and target modes and check whether you have found the counting rate maximum.
– In coupled scanning mode, move the target back by 10.2° (even if this takes you into the negative range!).
– Save the positions of the target and the sensor as the "zero position of the measuring system" by pressing TARGET, COUPLED and β LIMITS simultaneously.

Recording the diffraction spectrum:

– Start the software "X-ray Apparatus", check to make sure that the apparatus is connected correctly, and clear any existing measurement data using the button or the F4 key.
– Set the measuring time per angular step Δt = 10 s and the angular step width $\Delta\beta$ = 0.1°.
– Press the COUPLED key to activate 2ϑ coupling of target and sensor and set the lower limit of the target angle to 4° and the upper limit to 34°.
– Start measurement and data transfer to the PC by pressing the SCAN key.
– When you have finished measuring, save the measurement series under an appropriate name by pressing the button or the F2 key.

a) Bragg reflection at an NaCl monocrystal:

– Press the ZERO key to return the target and sensor to the current zero position.
– Remove the LiF crystal and carefully mount the NaCl crystal in its place.

Determining the zero position of the measuring system

– In coupled scanning mode, set the target to about 7.2° using the ADJUST knob.
– Switch on the tube high voltage with HV on/off.
– Leave the target position unchanged and, in sensor scanning mode, manually find the counting rate maximum for the first reflection maximum of the K_a line.
– Leave the sensor unchanged in the maximum counting-rate position and manually find the maximum of the counting rate in target mode.
– Switch between sensor and target modes and check whether you have found the counting rate maximum.
– In coupled scanning mode, move the target back by 7.2° (even if this takes you into the negative range!).
– Save the positions of the target and the sensor as the "zero position of the measuring system" by pressing TARGET, COUPLED and β LIMITS simultaneously.

Recording the diffraction spectrum:

– Start the software "X-ray Apparatus" or clear any existing measurement data using the button or the F4 key.
– Press the COUPLED key to activate 2ϑ coupling of target and sensor and set the lower limit of the target angle to 4° and the upper limit to 24°.
– Start measurement and data transfer to the PC by pressing the SCAN key.
– When you have finished measuring, save the measurement series under an appropriate name by pressing the button or the F2 key.

Measuring example

a) Bragg reflection at an LiF monocrystal:

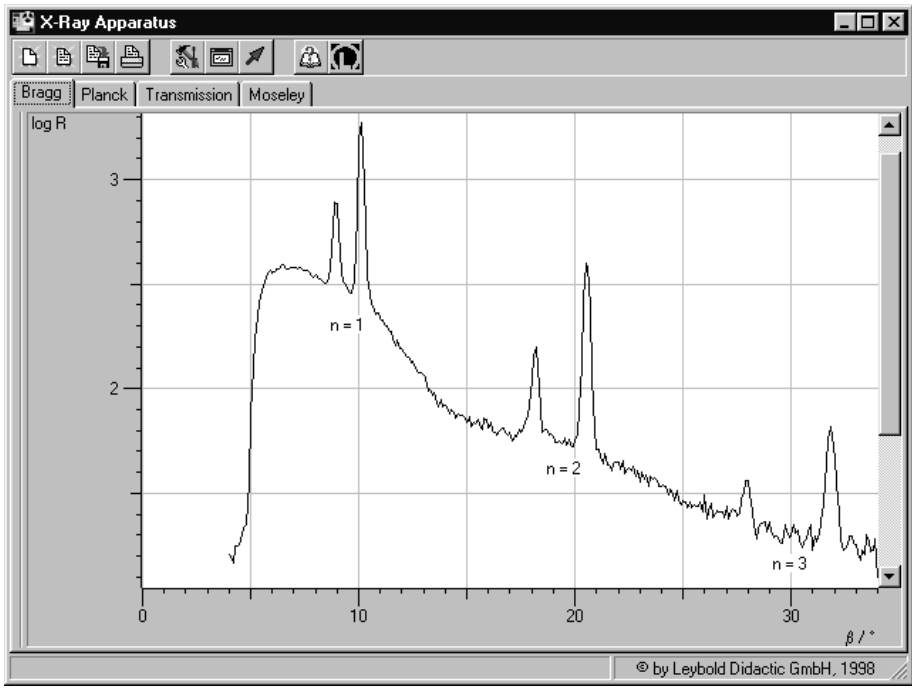


Fig. 4 Diffraction spectrum of x-rays in Bragg reflection to the third diffraction order at an LiF monocrystal with logarithmic display of counting rate *R*.
Parameters of x-ray tube:
U = 35 kV, *I* = 1 mA

b) Bragg reflection at an NaCl monocrystal:

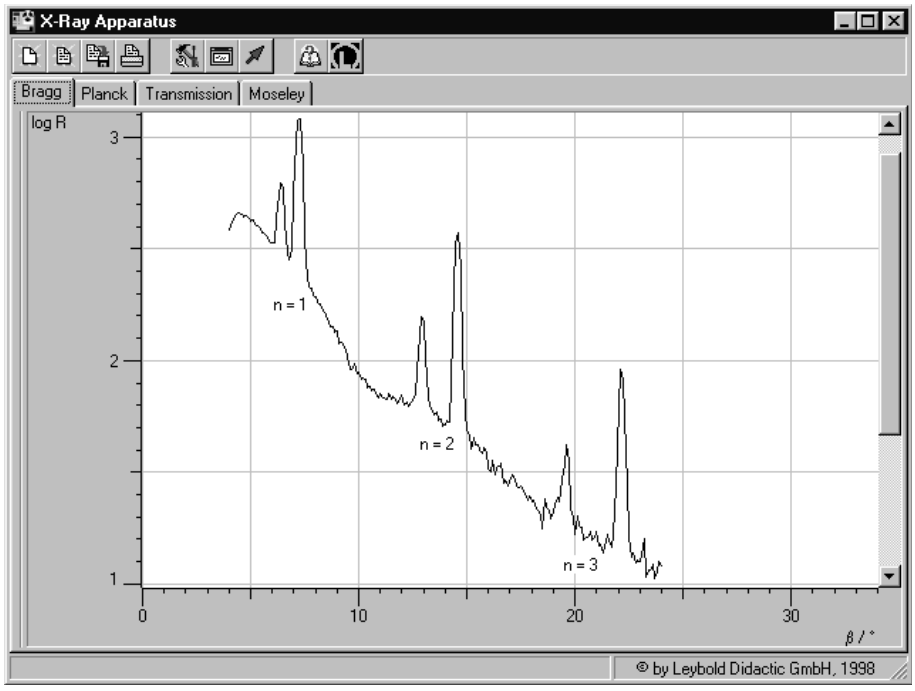


Fig. 5 Diffraction spectrum of x-rays in Bragg reflection to the third diffraction order at an NaCl monocrystal with logarithmic display of counting rate *R*.
Parameters of x-ray tube:
U = 35 kV, *I* = 1 mA

Evaluation

– In each diagram, click the right mouse button to access the evaluation functions of the software “X-ray Apparatus” and select the command “Calculate Peak Center” to evaluate the diffraction spectra.
– Using the left mouse button, mark the “full width” of each peak and write down the center values in a table as the glancing angle (see tables 2 and 3).
– For each glancing angle ϑ, calculate the values sin ϑ and and plot these value pairs in a diagram (see Fig. 6).

In each case, the results lie along a straight line through the origin; in accordance with (III), its slope corresponds to the lattice constant a_0.

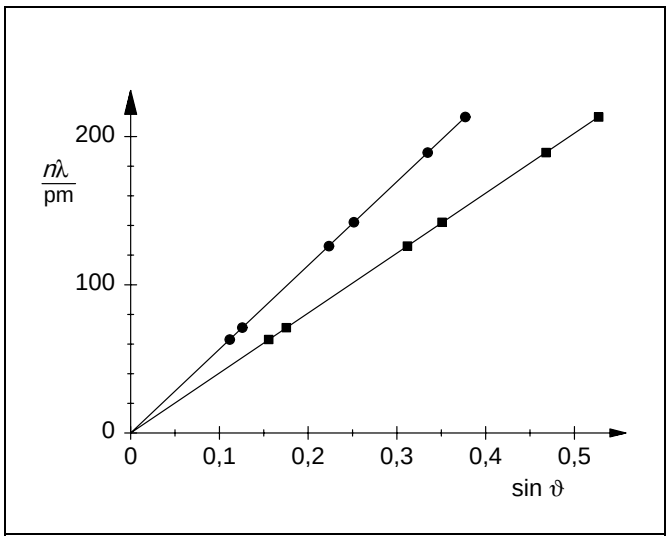


Fig. 6 Value pairs as a function of sin ϑ
LiF: squares, slope of line = 404.5 pm
NaCl: circles, slope of line = 565.2 pm

Table 2: Glancing angle ϑ of the LiF crystal

ϑ	sin ϑ	Line	*n*	$\frac{n \cdot \lambda}{\text{pm}}$
8.95°	0.156	K_β	1	63.06
10.10°	0.175	K_α	1	71.08
18.17°	0.312	K_β	2	126.12
20.54°	0.351	K_α	2	142.16
27.91°	0.468	K_β	3	189.18
31.82°	0.527	K_α	3	213.24

Table 3: Glancing angle ϑ of the NaCl crystal

ϑ	sin ϑ	Line	*n*	$\frac{n \cdot \lambda}{\text{pm}}$
6.41°	0.112	K_β	1	63.06
7.23°	0.126	K_α	1	71.08
12.91°	0.223	K_β	2	126.12
14.57°	0.252	K_α	2	142.16
19.55°	0.335	K_β	3	189.18
22.15°	0.377	K_α	3	213.24

Results

a) LiF crystal:

Measurement result: a_0 = 404.5 pm
Literature value [2]: a_0 = 402.7 pm
Ion radii [3]: 68 pm (Li^+), 133 pm (F^-)
Sum of ion radii: 201 pm

b) NaCl crystal:

Measurement result: a_0 = 565.2 pm
Literature value: a_0 = 564.02 pm
Ion radii [3]: 98 pm (Na^+), 181 pm (Cl^-)
Sum of ion radii: 279 pm

Conclusion: the LiF lattice shows a significantly smaller lattice constant than the NaCl lattice, as the radii of the ions involved are smaller.

Literature

[1] C. M. Lederer and V. S. Shirley, Table of Isotopes, 7th Edition, 1978, John Wiley & Sons, Inc., New York, USA.

[2] Handbook of Chemistry and Physics, 52nd Edition (1971–72), The Chemical Rubber Company, Cleveland, Ohio, USA.

[3] Charles Kittel, Introduction to Solid State Physics, John Wiley & Sons, Inc. New York, USA

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IISER Trivandrum
x-ray experiments

Experiment List:

14 P7.1.2.4 Debye-Scherrer Scan: determining the lattice plane spacing of polycrystalline powder samples

By Didactic Systems
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Debye-Scherrer Scan: determining the lattice-plane spacings of polycrystalline powder samples

Objects of the experiment

- Record the Debye-Scherrer spectrum of a polycrystalline sample.
- Calculate the lattice spacing from the result.

Principles

The Debye-Scherrer method is a way to determine the lattice constant of crystallised materials. A powdery crystalline sample is illuminated with monochromatic X-rays. The powder sample contains minute mono-crystals of about 5 - 50 μm diameter, so-called crystallites. A set of lattice planes in a crystallite leads to a reflection of the X-rays if it is aligned so that the Bragg condition:

$$n \cdot \lambda = 2 \cdot d \cdot \sin\vartheta \qquad \text{(I)}$$

n: diffraction order

λ: wave length

d: lattice plane spacing

ϑ: Bragg angle relative to the primary ray

is fulfilled (see Fig. 2 and experiment P6.3.3.1). This reflected beam is then either recorded on film (see Experiment P 7.1.2.3) or with the end-window counter tube . In general the crystallites are randomly oriented without any privileged direction so that there are always some crystallites in the crystal powder which correspond to a rotation of the crystallite under consideration around the primary axis. The angle between the diffraction reflection and the primary ray is 2ϑ (Fig 2).

In the arrangement of this experiment, the intensity of diffraction reflections will be recorded as function of the angle ϑ, where the counter tube passes through the circles of a conventional Debye-Scherrer photograph (Fig 1). The finer the power is, the more uniformly the individual reflections of the crystallites will be.

Cubic crystals:

If the consideration is restricted to cubic crystals the lattice plane spacing can be expressed in the form:

$$d = \frac{a_0}{\sqrt{h^2 + k^2 + l^2}} \qquad \text{(II)}$$

The integers h,k,l are the Miller indices of the set of lattice planes under consideration (see experiment P7.1.2.2). If equation (II) is inserted into equation (I), the quadratic form

$$\sin^2\vartheta = \left(\frac{\lambda}{2 \cdot a_0}\right)^2 \cdot \left((n \cdot h)^2 + (n \cdot k)^2 + (n \cdot l)^2 \cdot\right) \qquad \text{(III)}$$

is obtained (see experiment P7.1.2.3).

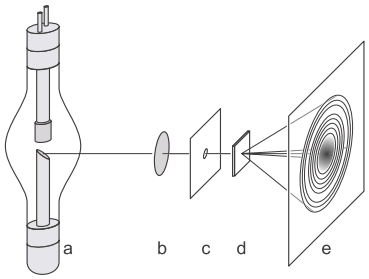


Fig. 1: Scheme of the classic setup for the Debye-Scherrer method. a: x-ray tube, b: Ni-filter, c: collimator, d: sample, e: screen.

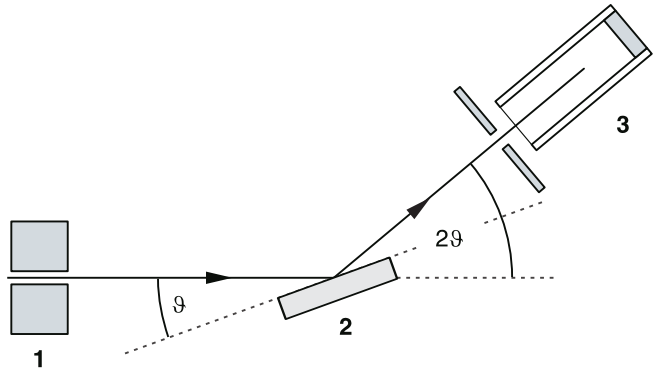


Fig. 2:Schematic diagram of diffraction of x-rays at a powder sample: 1: collimator, 2 powder sample, 3: counter tube.

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Formula (III) can be written in the short form

$$\sin^2 \vartheta = F \cdot Z \qquad \text{(V)}$$

with

$$F = \left(\frac{\lambda}{2 \cdot a_0} \right)^2 \qquad \text{(VI)}$$

and

$$Z = (n \cdot h)^2 + (n \cdot k)^2 + (n \cdot l)^2 \qquad \text{(VII)}.$$

Due to its components, Z is always an integer. For simple cubic crystals, every combination of integers n, h, k and l is allowed, however, the intensity of the diffraction reflections is weaker for higher diffraction orders n and greater *Miller* indices h, k, l.

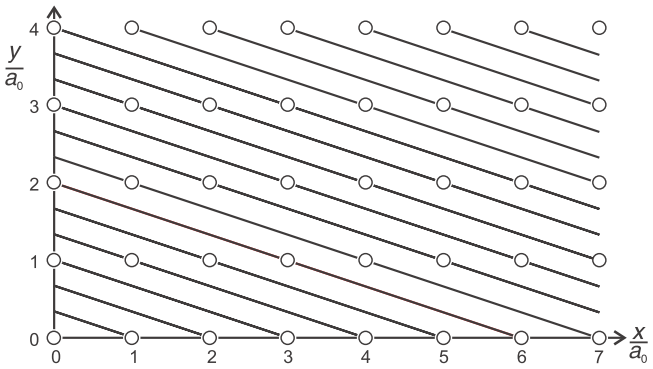


Fig 2: lattice planes with hkl = (130)

Crystals with NaCl structure:

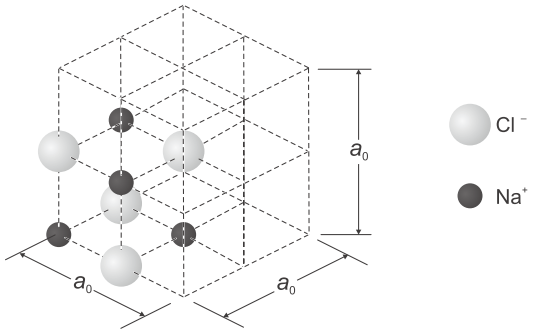


Fig. 3 Unit cell of an NaCl crystal

In the case of crystals with NaCl structure, the situation is more complicated because here alkali (e.g. Na) and halogenide (e.g. Cl) atoms take turns in a cubic lattice. The spatial lattice is no longer built up of simple lattice points with the distance a_0, but it is a series of cubic unit cells with an edge length a_0 (see Fig. 3). Every unit cell contains four alkali atoms with the co-ordinates

$$\boldsymbol{r}_1 = (0,0,0),\ \boldsymbol{r}_2 = \left(\frac{a_0}{2}, \frac{a_0}{2}, 0\right),\ \boldsymbol{r}_3 = \left(\frac{a_0}{2}, 0, \frac{a_0}{2}\right),\ \boldsymbol{r}_4 = \left(0, \frac{a_0}{2}, \frac{a_0}{2}\right)$$

and four halogenide atoms with the co-ordinates

$$\boldsymbol{r}_5 = \left(\frac{a_0}{2}, 0, 0\right),\ \boldsymbol{r}_6 = \left(0, \frac{a_0}{2}, 0\right),\ \boldsymbol{r}_7 = \left(0, 0, \frac{a_0}{2}\right),$$

$$\boldsymbol{r}_8 = \left(\frac{a_0}{2}, \frac{a_0}{2}, \frac{a_0}{2}\right).$$

At each atom of the unit cell the incoming X-ray is scattered, whereby the amplitudes of the scattered partial waves depend on the atomic number of the atom. The differences of path Δ_i of the partial waves can be calculated from the co-ordinates $\boldsymbol{r}_i$ of the atoms:

$$\Delta_i = (\boldsymbol{s}_1 - \boldsymbol{s}_2) \cdot \boldsymbol{r}_i \qquad \text{(VIII)}$$

$\boldsymbol{s}_1$: unit vector in the direction of the primary ray
$\boldsymbol{s}_2$: unit vector in the direction of the diffraction reflection

The partial waves scattered at the alkali atoms A and the halogenide atoms H interfere to form a common wave that is "scattered at the unit cell". The amplitude of this wave has the form

$$A = A_A + A_H \qquad \text{(IX)}$$

with

$$A_A = f_A \left(\cos\left(\frac{2\pi}{\lambda}\Delta_1\right) + \cos\left(\frac{2\pi}{\lambda}\Delta_2\right) + \cos\left(\frac{2\pi}{\lambda}\Delta_3\right) + \cos\left(\frac{2\pi}{\lambda}\Delta_4\right) \right)$$

und

$$A_H = f_H \cdot \left(\cos\left(\frac{2\pi}{\lambda}\Delta_5\right) + \cos\left(\frac{2\pi}{\lambda}\Delta_6\right) + \cos\left(\frac{2\pi}{\lambda}\Delta_7\right) + \cos\left(\frac{2\pi}{\lambda}\Delta_8\right) \right)$$

All waves that start from the unit cells interfere constructively if the *Bragg* condition (I) is fulfilled, which is equivalent to the *Laue* condition, which can be cast into the form

$$\boldsymbol{s}_1 - \boldsymbol{s}_2 = \lambda \cdot \boldsymbol{G} \ \text{ with } \ \boldsymbol{G} = (h,k,l) \cdot \frac{1}{a_0} \qquad \text{(X)}$$

for cubic crystals (see experiment P7.1.2.2) By inserting Eqs. (X) and (VIII) into (IX) one obtains

$$A_A = f_A \cdot (1 + \cos((h+k)\cdot\pi) + \cos((h+l)\cdot\pi) + \cos((k+l)\cdot\pi))$$

and

$$A_H = f_H \cdot (\cos(h\cdot\pi) + \cos(k\cdot\pi) + \cos(l\cdot\pi) + \cos((h+k+l)\cdot\pi)).$$

A short calculation shows that

$$A = \begin{cases} 4 \cdot f_A + 4 \cdot f_H, & \text{if } h, k \text{ and } l \text{ even} \\ 4 \cdot f_A - 4 \cdot f_H, & \text{if } h, k \text{ and } l \text{ odd} \\ 0, & \text{if } h, k \text{ and } l \text{ mixed} \end{cases} \qquad \text{(XI)}$$

The amplitudes A of the waves starting from the unit cells thus only are different form zero if all indices h, k, l are even or if they are all odd. A combination of even indices leads to a greater amplitude A than a combination of odd indices. For other crystal structures other selection rules apply.

Evaluating Debye-Scherrer scans

In this experiment, Debye-Scherrer scans of crystals with NaCl structure are taken. The Bragg angles for different h, k, l indices are recorded and compared to the calculated values.

There are two different ways to record the spectra with the x-ray apparatus:

- keeping the powder sample at a fixed angle and let only the sensor rotate
- moving powder sample and sensor in a coupled motion, similar to the ordinary single crystal scans

Rotating a powder sample should not make any difference at all, but for geometrical reasons of the non-circular sample the second way is preferred. Also, rotating the sensor alone would give the angle 2ϑ, in contrast to coupled rotation, where the X-ray apparatus displays the angle ϑ.

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Setup

Safety notes

The X-ray apparatus fulfils all regulations on the design of an X-ray apparatus and fully protected device for instructional use and is type approved for school use in Germany (BfS 05/07 V/Sch RöV).

The built-in protective and shielding fixtures reduce the dose rate outside the X-ray apparatus to less than 1 µSv/h, which is of the order of magnitude of the natural background radiation.

■ Before putting the X-ray apparatus into operation, inspect it for damage

■ Protect the X-ray apparatus against access by unauthorized persons.

■ When switching the X-ray apparatus on, check whether the ventilator in the tube chamber starts rotating to avoid overheating of the X-ray tube.

■ For further details see instruction sheet of the X-ray apparatus.

Apparatus

1 X-ray apparatus , basic device..... 554 800
1 X-ray tube Cu..... 554 862
1 Goniometer..... 554 831
1 End-window counter with cable 559 01
1 Set of 2 crystal powder holders..... 554 842
1 NaCl, 250 g..... 673 5700
1 Spatula 666 960
1 Pestle, 100 mm long 667 091
1 Mortar, porcelain, 70 mm Ø 667 092

additionally suggested:

e.g. KCl or LiF powder

1 PC with Windows 98/2000/XP/Vista

- Mount the Goniometer in the experiment chamber of the X-ray apparatus. To bring the counter tube out of stray radiation, put the Goniometer to the very left position and use the full length of the sensors arm. See Fig 4
- Mount the Ni-filter on the collimator to produce monochromatic copper Kα X-rays. The filter must be mounted on the left side on the collimator, facing the tube, not on the right hand side, where it would produce a lot of stray radiation into the experiment chamber.
- Prepare a powder sample by using the mortar and the pestle. The NaCl (or other) salt is grind in a mortar until no grains can be felt when rubbing the powder between the finger tips. Do not try to prepare to much powder at once, as it will take a lot of time. An amount of 2 times the volume of the powder holder will take approximately 5 minutes to pulverise.

Note: Coarse powder samples yield Debye-Scherrer reflections of inhomogeneous intensity distributions.

- The powder holder consists of one aluminum frame and a solid aluminum plate. Put the frame onto the plate and fill the powder into the frame, move both aluminum parts together close to the sample holder inside the X-ray apparatus and slide the frame sideways onto the sample holder, so that the powder is always supported either by the aluminum or by the plastic parts.



Fig 4: The experiment chamber

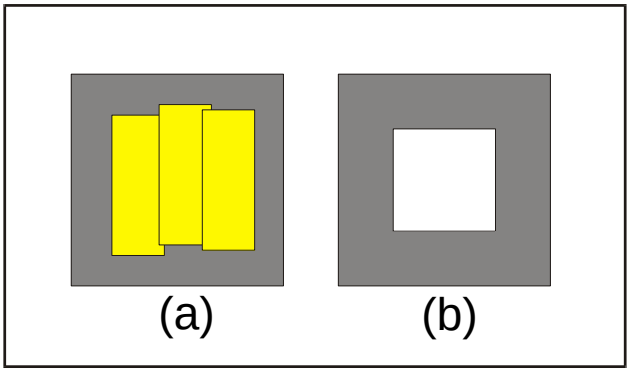


Fig. 5: Preparation of the powder holder frame schematically:
(a) rear side with strips of scotch-tape
(b) front side with squeezed powder

Or (preferred):

- Attach strips of scotch-tape to the rear of the powder holder and fill the prepared powder into the frame (Fig. 5).
- Squeeze the powder into the frame with the spatula as dense as possible.

Note: Instead of using the crystal powder holder a self-made styrofoam block can be used, where the powder is pressed into a recess. This will lower the background radiation even more.

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Carrying out the experiment

- Start the software and the x-ray apparatus (see “Recording a diffraction spectrum” in experiment P7.1.2.1)
- Set the tube high voltage to U = 35.0 kV and the emission current I = 1.0 mA.
- Set the time per angular step Δt = 10 s and the angular step width $\Delta\beta$ = 0.1°.
- Select “coupled”, set appropriate start and stop angles β for the scan. Usually 10 to 40 deg is sufficient, but up to 60 ° gives additional lines.
- Switch on the high voltage and press the automatic scan button for recording the spectra.
- To check for the existence of the weak, but important <111> reflection, scan that region left of the 200 reflex (11°-18° for NaCl) with Δt = 30 s at least.

Measuring example

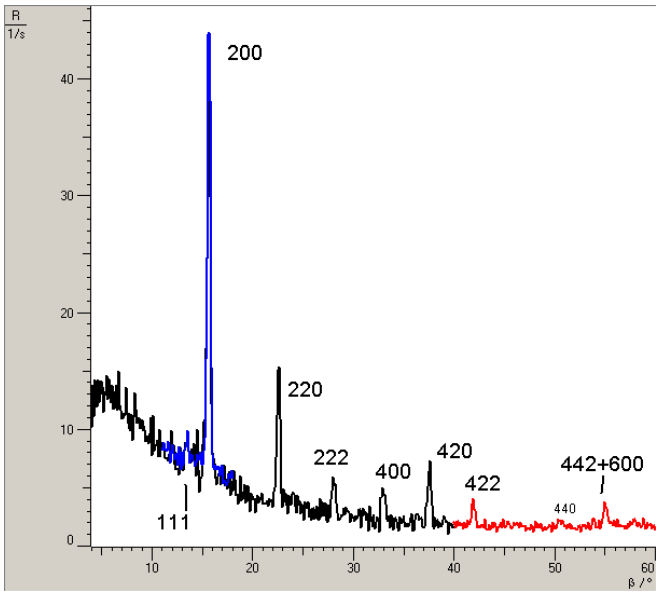


Fig. 6: Spectrum of a powdered NaCl sample recorded with monochromatic copper Kα x-rays. Black: 10s per point, red + blue: 30 s

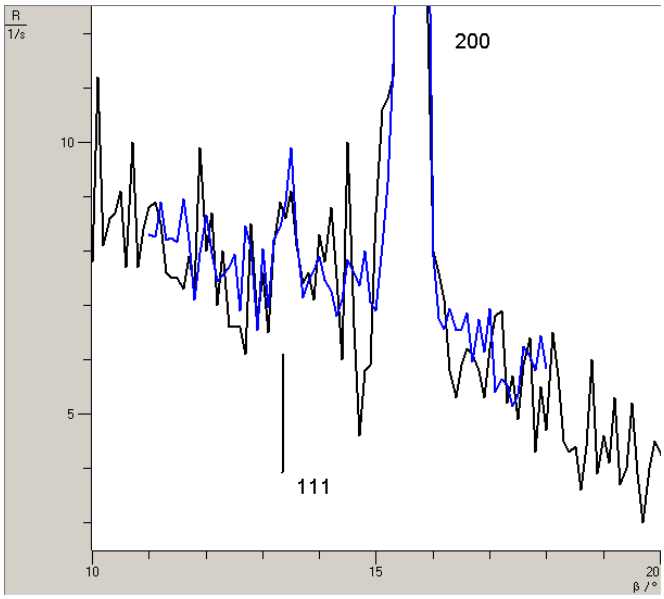


Fig. 7: Detail of the hkl = 111 reflection.

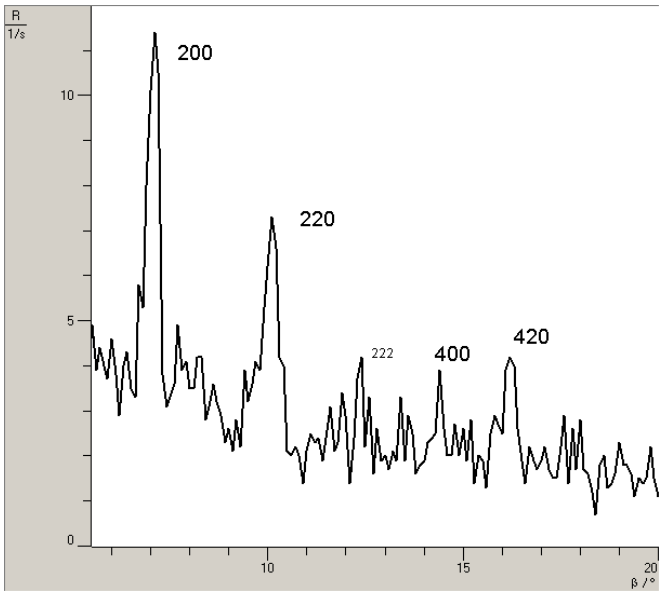


Fig. 8: Spectrum of a powdered NaCl sample recorded with monochromatic molybdenum Kα x-rays. 10s per point.

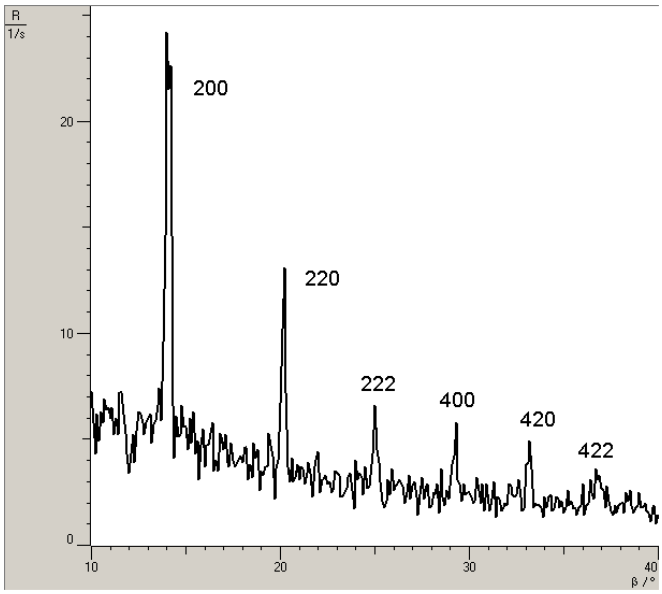


Fig. 9: Spectrum of a powdered KCl sample recorded with monochromatic copper Kα x-rays. 10s per point.

Evaluation and results

The spectra show several different peaks, which correspond to the circles in a Debye-Scherrer photograph. In the measurement examples shown, these peaks are already labelled with the appropriate hkl numbers.

Several combinations of anode materials and crystal powders are shown in the measurement examples. Figure 6 and 7 show the experiment described above, copper Kα with NaCl powder, which will be evaluated in detail below.

Figure 8 shows the same powder, but using a molybdenum anode, which is also possible. The angles of reflection scale according to the wavelength, but note that the amplitude of the reflexes is much smaller than using a copper tube, as the shorter wavelength X-rays are scattered less by the given amount of powder.

Figure 9 shows the copper tube with a different powder, KCl, showing a different lattice constant. Note here, that there is

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absolutely no trace of the hkl= 111 reflection. This is due to the fact that both atoms (K and Cl) show the same electronic configuration (Argon) in the crystal and therefore $f_A = f_H$. According to equation XI, the hkl all odd reflexes vanish in this case.

For a detailed analysis one can either start with the literature values for the lattice constant a0, calculate all possible hkl combinations according to equation III and match the peaks recorded to the calculated values.

The other way to do the analysis is shown in experiment P7.1.2.3, starting with an unknown lattice constant and guessing the hkl numbers so that the factor F (Equation VI) is constant.

The 111 reflection in figure 7 is of specific importance in determining the crystal structure. According to equation XI, all the reflections from a NaCl crystal are either all even number or all odd numbers. If this reflection was not present, one could also assume a simple cubic crystal with half the lattice constant.

Detailed evaluation

The literature values for different lattice constants are:

Crystal	a0/pm
LiF	402,8
NaCl	564,02
KCl	629,3
KBr	659,66

The wavelength of the Ka line is 154,2 pm for a copper anode and 71,08 pm for a molybdenum anode.

According to equation VI the combination Cu-Anode, NaCl powder yields F= 0,01869.

Choosing different combinations of hkl, it is now possible to calculate Z from equation VII and finally calculate the angle ϑ from equation V.

h	k	l	Z	ϑ
1	1	1	3	13,70
2	0	0	4	15,87
2	2	0	8	22,75
3	1	1	11	26,96
2	2	2	12	28,26
4	0	0	16	33,15
3	3	1	19	36,57
4	2	0	20	37,69
4	2	2	24	42,04
3	3	3	27	45,26
5	1	1	27	45,26
4	4	0	32	50,65
5	3	1	35	53,97
4	4	2	36	55,10
6	0	0	36	55,10
6	2	0	40	59,83
5	3	3	43	63,69
6	2	2	44	65,06

Angles marked in blue are from the all odd hkl reflection, while the black ones are from the all even ones. According to equation XI, the odd reflections are low in intensity ($f_A - f_H$.), while the even reflections are of high intensity ($f_A + f_H$.).

Comparing the calculated values to figure 6, the peaks can be identified and named with the correct hkl combination, verifying that the structure of the NaCl crystal is as it was assumed.

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